Linux Kernel Compilation

Operating Systems

CS3015 - 2026I
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Executive Summary

- **Motivation:** Understand the Linux Kernel ☺
- **Problem:** Change from educational to development environments.
- **Overview:**

- Create a config
- Compile Linux Kernel
- Adding to Grub

- **Conclusion:** I can start my Kernel journey ☺

Outline

Working Environment

Compile the kernel

GRUB

Conclusions

Working environment

1. **Create a Virtual Machine using Virtualbox with Ubuntu 20.04 LTS (or a new LTS version)**.
   - **If possible use half o your available cores. 15 GB virtual disk size.**
     - https://www.virtualbox.org/wiki/Downloads
     - https://ubuntu.com/download/desktop

1. **Once created, install Virtualbox Guest Additions**.
2. **Install the following packages**:

```
sudo apt update

sudo apt install git fakeroot build-essential tar libncurses-dev tar xz-utils
libssl-dev bc stress python3-distutils libelf-dev linux-headers-$(uname -r)
bison flex libncurses5-dev util-linux net-tools linux-tools-$(uname -r)
exuberant-ctags cscope sysfsutils gnome-system-monitor curl perf-tools-
unstable gnuplot rt-tests indent tree smem libnuma-dev numactl hwloc bpfcc-
tools sparse flawfinder cppcheck tuna trace-cmd virt-what
```

3. **Download the Linux Kernel 5.4 longterm source (or a new longterm source version):**

   https://cdn.kernel.org/pub/linux/kernel/v5.x/linux-5.4.161.tar.xz

Kernel Source

- Download the source:

```
mkdir kernel; cd kernel
wget https://cdn.kernel.org/pub/linux/kernel/v5.x/linux-5.4.161.tar.xz
tar -xvf linux-5.4.161.tar.xz
export LLKD_KSRC=${HOME}/kernel/linux-5.4.161/
```

- Checking the source:

```
jorge@sican-vm:~/kernel/linux-5.4.161$ ls
arch          crypto        ipc       MAINTAINERS      samples   virt
block         Documentation Kbuild    Makefile         scripts
certs         drivers       Kconfig   mm               security
commands.txt  fs            kernel    modules.builtin  sound
COPYING       include       lib       net              tools
CREDITS       init          LICENSES  README           usr
jorge@sican-vm:~/kernel/linux-5.4.161$
```

Understanding the Makefile


```
> uname -r
5.11.0-40-generic

> less Makefile
  # SPDX-License-Identifier: GPL-2.0
  VERSION = 5
  PATCHLEVEL = 4
  SUBLEVEL = 161
  EXTRAVERSION =
  NAME = Kleptomaniac Octopus

  # *DOCUMENTATION*
  # To see a list of typical targets execute "make help"
  # More info can be located in …
```

Release # component	Meaning	Example numbers
Major # (or w)	Main or major number; currently, we are on the 5.x kernel series, thus the major number is 5.	2, 3, 4, and 5
Minor # (or x)	The minor number, hierarchically under the major number.	0 onward
[patchlevel] (or y)	Hierarchically under the minor number – also called the ABI or revision – applied on occasion to the stable kernel when significant bug/security fixes are required.	0 onward
[-EXTRAVERSION] (or -z)	Also called localversion; typically used by distribution kernels to track their internal changes.	Varies; Ubuntu uses w.x.y-'n'-generic

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Create config

- **Method 1:** Based on templates

```
ls ${LLKD_KSRC}/arch/
ls ${LLKD_KSRC}/arch/ia64/configs/
```

- **Method 2:** Manual copy of actual system config.

```
cp /boot/${uname -r} ${LLKD_KSRC}/.config
```

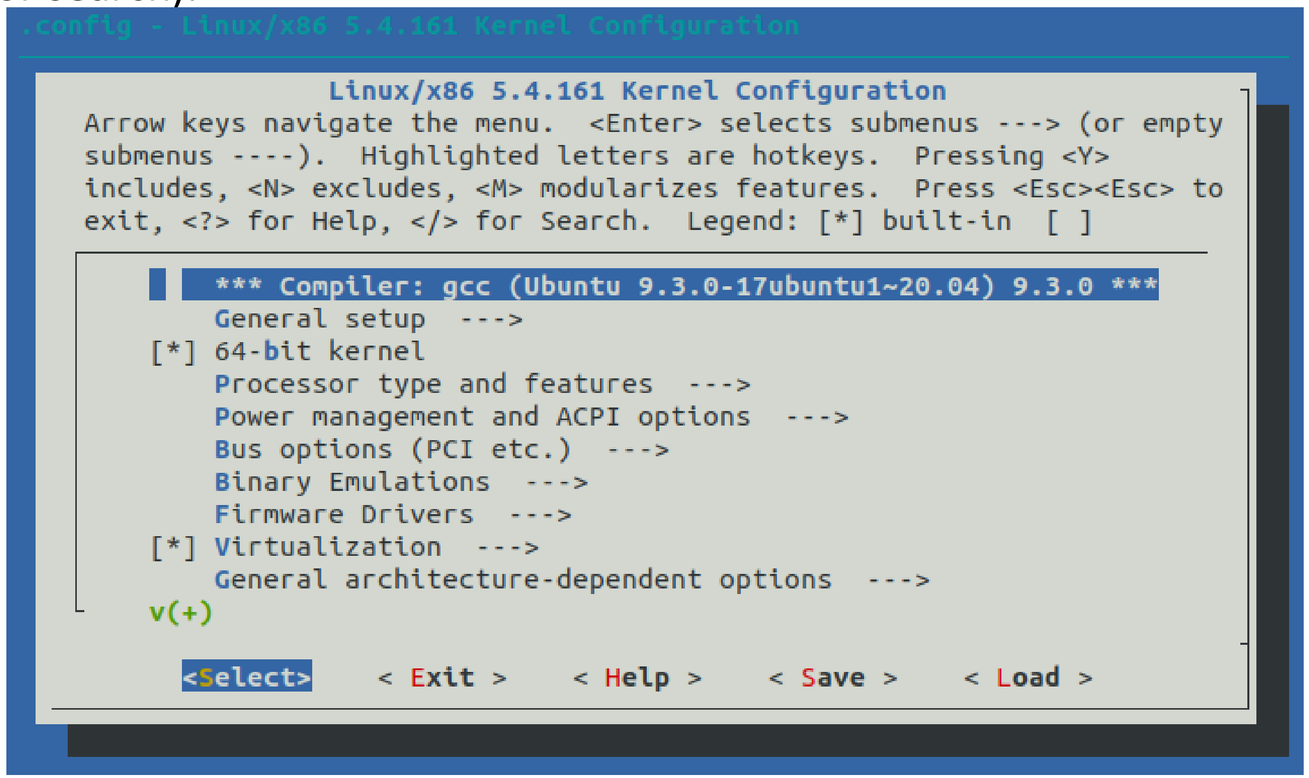
- **Method 3:** Automatic copy of actual system config.

```
lsmod > /tmp/lsmod.now
cd ${LLKD_KSRC}
make LSMOD=/tmp/lsmod.now localmodconfig
diff .config .config.old
```

Method 4 (recommended)> Configure with UI

- Start the UI for configuration (press / for search):

```
make menuconfig
```



- **Recommended:** Try to disable CONFIG_DEBUG_INFO_BTF. Search for BTF (debug kernel compile)
- Exit and save your config

Compilation

- Optional: Check your changes on the configuration:

```
grep IKCONFIG .config
```

- Compile using all the possible cores (e.g.: with 2 cores):

```
>time make -j2
LD [M]  fs/nls/nls_iso8859-1.ko
  LD [M]  sound/ac97_bus.ko
  LD [M]  kernel/configs.ko
  LD [M]  sound/core/seq/snd-seq.ko
  LD [M]  sound/core/snd-pcm.ko
  LD [M]  sound/core/snd-seq-device.ko
  LD [M]  sound/core/snd-timer.ko
  LD [M]  sound/core/snd.ko
  LD [M]  sound/pci/ac97/snd-ac97-codec.ko
  LD [M]  sound/pci/snd-intel8x0.ko

real       18m58,057s
user       32m43,007s
sys        3m56,452s
```

Modules

- Find your modules:

```
$ cd ${LLKD_KSRC}
$ find . -name "*.ko" ./arch/x86/events/intel/intel-rapl-perf.ko
./arch/x86/crypto/crc32-pclmul.ko ./arch/x86/crypto/ghash-clmulni-
intel.ko [...]
./net/ipv4/netfilter/ip_tables.ko ./net/sched/sch_fq_codel.ko
$ find . -name "*.ko" | wc -l
59


$ find . -name "*.ko" -ls | egrep -i "vbox|msdos|uio" | awk
'{printf "%-40s %9d\n", $11, $7}' ./fs/fat/msdos.ko 361896
./drivers/virt/vboxguest/vboxguest.ko 948752
./drivers/gpu/drm/vboxvideo/vboxvideo.ko 3279528
./drivers/uio/uio.ko 408136 ./drivers/uio/uio_pdrv_genirq.ko 324568
$
```

- Install modules:

```
sudo make modules_install
```

TODO

- **Challenge:** complete module installation and add your kernel to GRUB (see following slides).

Check your modules

```
  INSTALL drivers/powercap/intel_rapl_common.ko
  INSTALL drivers/powercap/intel_rapl_msr.ko
  INSTALL drivers/video/fbdev/core/fb_sys_fops.ko
  INSTALL drivers/video/fbdev/core/syscopyarea.ko
  INSTALL drivers/video/fbdev/core/sysfillrect.ko
  INSTALL drivers/video/fbdev/core/sysimgblt.ko
  INSTALL drivers/virt/vboxguest/vboxguest.ko
  INSTALL drivers/xen/xen-privcmd.ko
  INSTALL fs/autofs/autofs4.ko
  INSTALL fs/nls/nls_iso8859-1.ko
  INSTALL kernel/configs.ko
  INSTALL sound/ac97_bus.ko
  INSTALL sound/core/seq/snd-seq.ko
  INSTALL sound/core/snd-pcm.ko
  INSTALL sound/core/snd-seq-device.ko
  INSTALL sound/core/snd-timer.ko
  INSTALL sound/core/snd.ko
  INSTALL sound/pci/ac97/snd-ac97-codec.ko
  INSTALL sound/pci/snd-intel8x0.ko
  DEPMOD  5.4.161
jorge@sican-vm:~/kernel/linux-5.4.161$ ls /lib/mod
modprobe.d/     modules/        modules-load.d/
jorge@sican-vm:~/kernel/linux-5.4.161$ ls /lib/modules
5.11.0-27-generic  5.11.0-40-generic  5.4.161
```

Your kernel (with modules) name

Installed to the system

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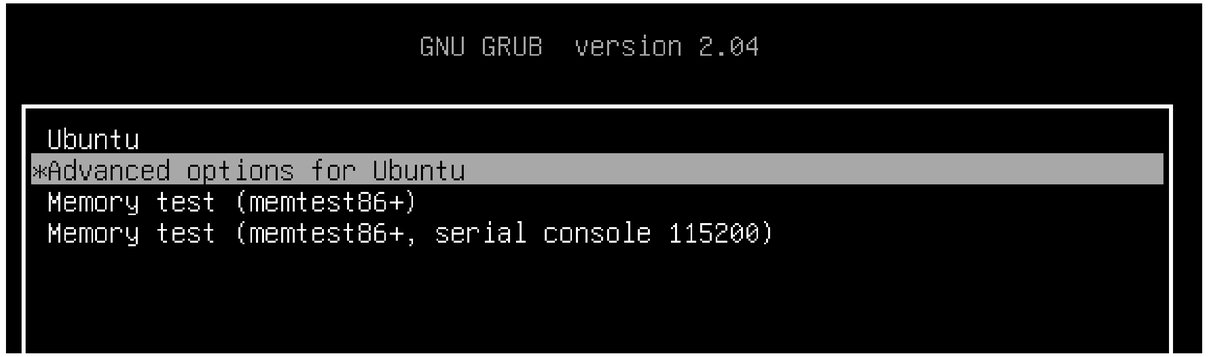
Add to GRUB

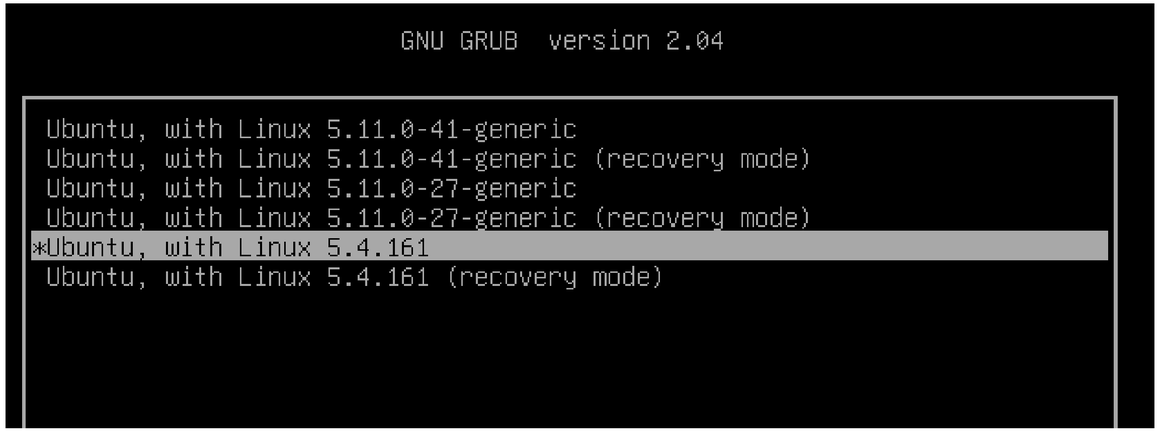
```
jorge@sican-vm:~/kernel/linux-5.4.161$ sudo make install

sh ./arch/x86/boot/install.sh 5.4.161 arch/x86/boot/bzImage \
    System.map "/boot"
run-parts: executing /etc/kernel/postinst.d/apt-auto-removal 5.4.161 /boot/vmlinuz-5.4.161
run-parts: executing /etc/kernel/postinst.d/initramfs-tools 5.4.161 /boot/vmlinuz-5.4.161
update-initramfs: Generating /boot/initrd.img-5.4.161
run-parts: executing /etc/kernel/postinst.d/unattended-upgrades 5.4.161 /boot/vmlinuz-5.4.161
run-parts: executing /etc/kernel/postinst.d/update-notifier 5.4.161 /boot/vmlinuz-5.4.161
run-parts: executing /etc/kernel/postinst.d/vboxadd 5.4.161 /boot/vmlinuz-5.4.161
VirtualBox Guest Additions: Building the modules for kernel 5.4.161.

VirtualBox Guest Additions: Look at /var/log/vboxadd-setup.log to find out what
went wrong
run-parts: executing /etc/kernel/postinst.d/xx-update-initrd-links 5.4.161 /boot/vmlinuz-5.4.161
I: /boot/initrd.img.old is now a symlink to initrd.img-5.11.0-40-generic
I: /boot/initrd.img is now a symlink to initrd.img-5.4.161
run-parts: executing /etc/kernel/postinst.d/zz-update-grub 5.4.161 /boot/vmlinuz-5.4.161
Sourcing file `/etc/default/grub'
Sourcing file `/etc/default/grub.d/init-select.cfg'
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-5.11.0-40-generic
Found initrd image: /boot/initrd.img-5.11.0-40-generic
Found linux image: /boot/vmlinuz-5.11.0-27-generic
Found initrd image: /boot/initrd.img-5.11.0-27-generic
Found linux image: /boot/vmlinuz-5.4.161
Found initrd image: /boot/initrd.img-5.4.161
Found memtest86+ image: /boot/memtest86+.elf
Found memtest86+ image: /boot/memtest86+.bin
done
```

Installation to Grub completed

Booting with your kernel

- Restart. **Press ESC during boot,** and select your kernel.





- **Check your kernel version.**

```
jorge@sican-vm:~$ uname -a
Linux sican-vm 5.4.161 #1 SMP Tue Nov 30 22:41:27 -05 2021 x86_64 x86_64 x86_64
GNU/Linux
jorge@sican-vm:~$ uname -r
5.4.161
jorge@sican-vm:~$
```

Congratulations! ☺

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Conclusions

- **We are ready to begin our Linux Kernel journey.**

- **Happy hacking!☺**

#TuxTurns30

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Questions

1. Explain your config file, highlight only the important features. Have you used your current maching config or created a custom file?
2. Measure the kernel compilation time.
3. How many modules have you compiled?
4. Show evidence of your kernel compilation and installation. Include screenshots.
5. Create a bash script with all your compilation commands.

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